Does Monetary Tightening Improve Banking Stability?

The Role of Bank Cost Efficiency

Presentation prepared for Bank of Ghana

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Roadmap

1. Motivation and research question
2. Data, measurement, and identification
3. Core empirical findings
4. Dynamic transmission channels
5. DSGE interpretation and counterfactuals
6. Policy takeaways for central banks

Why this question matters?

Central question

Does a monetary tightening **improve** banking stability, or does it **weaken** banks once borrower stress builds up?

Main answer in one line

A policy tightening tends to **raise bank stability on impact**, but that gain can **erode in the medium run**; the reversal is **smoother for more cost-efficient banks**.

The paper argues that both views can be right — but **at different horizons**.

The missing conditioning variable is **bank cost efficiency**: operational efficiency proxies for monitoring, screening, repricing, and loss-mitigation capacity.

This matters for monetary authorities because the banking response to tightening is not purely a capital or liquidity story.

What is new in this paper?

Data and measurement

Global panel of **3,903 banks**

95 countries, annual data, **1996–2024**

Bank cost efficiency recovered from a **stochastic translog cost frontier**

Cross-country comparability achieved with a **stochastic metafrontier**

Identification and interpretation

Monetary shocks measured from **Taylor-rule deviations** using official and hybrid policy stances

Evidence combined from **fixed effects**, **IV**, and **local projections**

A bank-heterogeneous **New Keynesian DSGE** model rationalises the sign reversal and efficiency heterogeneity

Data Snapshot and Variable Construction

Main stability metric: **Z-score** (distance to insolvency)

Complementary risk metric: **NPL growth**

Policy variable: standardised monetary stance, constructed as deviation from a Taylor-type benchmark

Average values in the full sample:

- Z-score: **26.47**
- Global metafrontier cost efficiency: **0.45**
- Bank liquidity ratio: **35.61%**
- GDP growth: **2.92%**
- CPI inflation: **3.79%**

Variable	Symbol	Obs	Mean	Std. Dev.
Panel A: Stochastic metafrontier variables				
Total cost (log)	lcost	49,859	17.36	3.61
Output: Gross Loans (log)	ly1	49,859	15.92	3.66
Price of Capital Input scaled by price of labour (log)	lx2: ln(w2/w1)	49,859	3.9773	1.413
Price of borrowed funds scaled by price of labour (log)	lx3: ln(w2/w1)	49,859	0.20	1.451
GDP per capita growth	GDP p.c. growth	49,859	1.95	3.49
Inflation (CPI) (%)	Inflation	49,859	4.4279	10.8084
Inflation (GDP deflator) (%)	Inflation	49,859	4.633	12.43
Panel B: Efficiency scores				
Cost efficiency (Country frontier)	$\hat{CE}^d$	44,652	0.68	0.21
Cost efficiency (Global metafrontier)	$M\hat{C}E$:(CostEff)-Global	44,652	0.45	0.18
Cost efficiency (Regional metafrontier)	$M\hat{C}E$:(CostEff)-Regional	44,652	0.53	0.22
Cost efficiency (Development-status metafrontier)	$M\hat{C}E$:(CostEff)-Devstat	44,652	0.47	0.19
Cost efficiency (Income-dynamics metafrontier)	$M\hat{C}E$:(CostEff)-IncGrp	44,652	0.49	0.19
Cost efficiency (ECB group metafrontier)	$M\hat{C}E$:(CostEff)-ECBM	44,652	0.46	0.19
Panel C: Competition (Lerner index)				
Lerner index (Country frontier)	$\hat{L}^{(o)}_{it}$	40,562	0.52	0.25
Lerner index (Global metafrontier)	$\hat{L}^{M}_{it}$	40,562	0.67	0.21
Panel D: Other variables				
Monetary policy stance (official, standardised)	Policy^s_j	44,269	-0.004	0.96
Monetary policy stance (hybrid, standardised)	Policy^s_j	44,269	-0.003	0.95
Bank stability: (Z-score)	Z-score	44,652	26.47	23.65
Bank stability: NPL growth (log change)	$\Delta \ln(\text{NPL})$	39,227	0.13	1.14
Bank stability: NPL growth minus loan growth	$\Delta \log(\text{NPL}) - \Delta\ell$	39,177	-12.12	3.71
Real GDP (constant 2015, USD billions)	Real GDP	44,652	6866.95	7735.67
Liquidity ratio (%)	Bank Liquidity	44,652	35.61	44.40
Bank size (log of assets)	Size	44,652	16.65	3.56
Asset structure (%)	Asset Structure	44,652	1.72	3.72
Bank concentration (%)	Bank concentration	44,533	42.69	19.72
GDP growth (%)	GDP growth	44,652	2.92	3.20
Inflation (CPI) (%)	Inflation	44,652	3.79	5.11
Institutional Quality	Quality	42,760	0.59	0.85
Net interest margin (%)	NIM	44,649	3.8290	3.6209
Loan growth (log change)	$\Delta\ell$	40,569	12.13	3.59
Loans-to-assets ratio (%)	Loans/TA	44,595	59.52	20.29
Macroprudential: Liquidity	Liquidity	43,816	0.20	0.68
Macroprudential: FX-related limit (LFX)	LFX	43,816	0.02	0.24
Macroprudential: Loan-to-value limit (LTV)	LTV	43,816	0.02	0.40
IV: Central bank independence	CBI	44,188	0.52	0.18

How cost efficiency is measured?

Stage 1: stochastic frontier

The paper estimates a **translog cost frontier** using total cost, output, and input prices to recover bank-level inefficiency.

Stage 2: stochastic metafrontier

Country-specific frontiers are then mapped into broader technology sets — global, regional, development-status, income-group, and ECB-group metafrontiers — so that efficiency is **comparable across heterogeneous banking systems**.

Interpretation: efficiency is not just a profitability metric.

It proxies for **operational capability** in monitoring, repricing, provisioning, and absorbing credit losses.

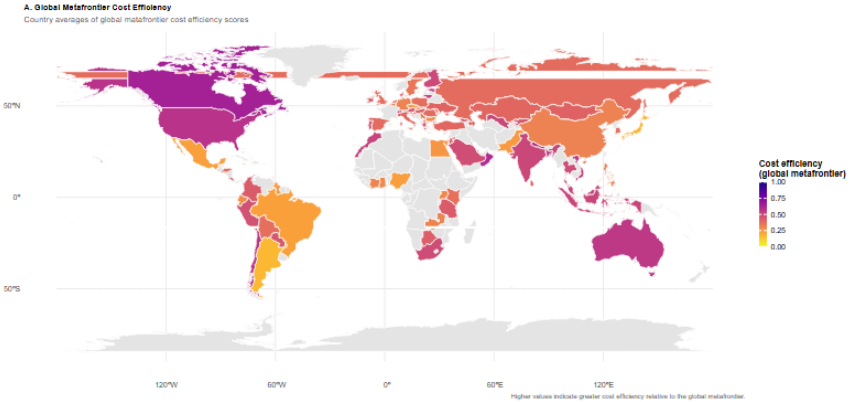
$$\text{Policy}^{z}_{j,t} = i_{j,t} - \widehat{i}^{Taylor}_{j,t}$$

$$Z\text{-score}_{i,j,t} = \frac{ROA + CAR}{\sigma(ROA)}$$

Economic intuition

Higher cost efficiency $\Rightarrow$ stronger risk-control response to tighter policy.

Figure 1: Global distribution of cost efficiency

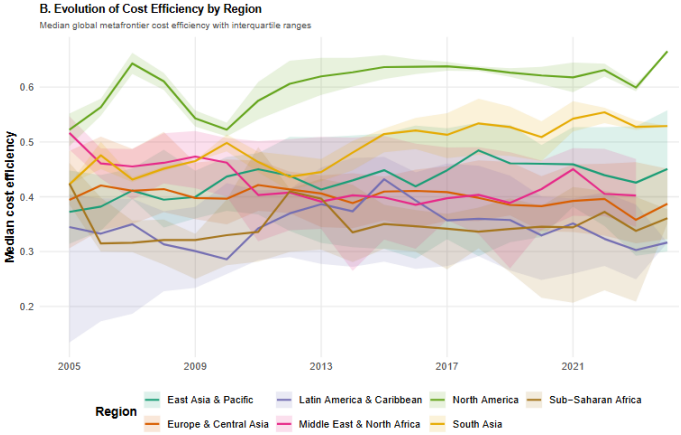


Note: Grey areas represent missing data.

Cost efficiency is **far from uniform** across countries.

The cross-country spread motivates treating efficiency as a state variable that can shape policy transmission.

Figure 2. Cost efficiency also varies over time and across regions



Regional medians move differently over time, indicating persistent technology and management gaps.

A single "representative bank" would hide substantial heterogeneity in resilience.

Identification Strategy

Baseline reduced-form design

$$Z_{i,j,t} = \alpha_i + \lambda_t + \eta_j + \beta\, Policy^z_{j,t-1} + \Gamma' X_{i,j,t-1} + \varepsilon_{i,j,t}$$

Bank, year, and country fixed effects

Official and hybrid monetary-policy measures

Controls for liquidity, size, asset structure, concentration, growth, inflation, and institutional quality

Robustness layers

IV: central bank independence as an instrument for the policy stance

Local projections: dynamic response path of Z-score and NIM

Macroprudential controls: liquidity tools, LFX, and LTV restrictions

Why this matters

A positive average coefficient does *not* imply monotone effects over time. The dynamic path is a core object in the paper.

Baseline Regression Results

Policy variable:	Hybrid		Official	
Model:	(1)	(2)	(3)	(4)
$\text{Policy}^{z}_{j,t-1}$	0.1391***	0.1159***	0.1736***	0.1544***
	(0.0389)	(0.0374)	(0.0395)	(0.0381)
Cost efficiency	4.301***	2.621***	4.297***	2.610***
	(0.8219)	(0.7868)	(0.8221)	(0.7872)
Bank liquidity	0.0298***	0.0161***	0.0298***	0.0161***
	(0.0030)	(0.0022)	(0.0030)	(0.0022)
Size	-2.754***	-1.855***	-2.752***	-1.852***
	(0.1980)	(0.1781)	(0.1981)	(0.1781)
Asset structure	0.1087***	0.0576	0.1089***	0.0575
	(0.0405)	(0.0384)	(0.0404)	(0.0383)
Bank Concentration	0.0272**	0.0440***	0.0275**	0.0444***
	(0.0109)	(0.0121)	(0.0109)	(0.0121)
GDP growth	0.0413***	0.0352**	0.0382**	0.0325**
	(0.0153)	(0.0142)	(0.0154)	(0.0142)
Inflation (CPI)	0.0362***	0.0089	0.0357***	0.0086
	(0.0116)	(0.0109)	(0.0116)	(0.0109)
Institutional Quality	2.912***	3.246***	2.849***	3.175***
	(0.7138)	(0.7253)	(0.7125)	(0.7240)
Fixed-effects				
Bank	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Country	Yes	Yes	Yes	Yes
No. of Banks	3,903	3,773	3,903	3,773
N	42,519	39,170	42,519	39,170
R^2	0.95	0.96	0.95	0.96

Baseline FE Results: tightening raises stability on average

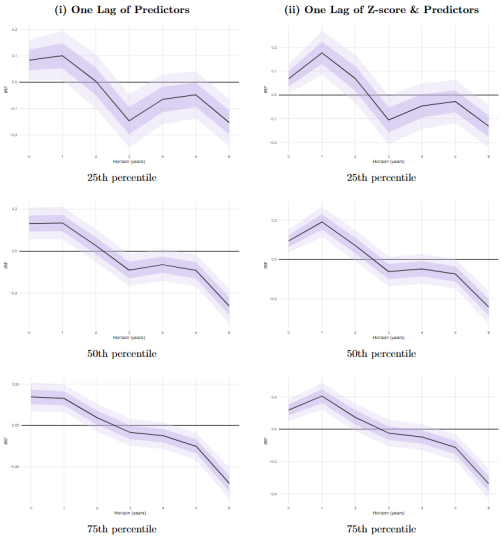
	Hybrid policy		Official policy	
Specification	Contemp.	Lagged ctrls	Contemp.	Lagged ctrls
Baseline FE	**0.1391*** **	**0.1159*** **	**0.1736*** **	**0.1544*** **
+ Liquidity & LFX	0.1367***	0.1057***	0.1724***	0.1435***
+ Liquidity & LTV	0.1243***	0.0942**	0.1612***	0.1332***

Across all FE specifications, a tighter monetary stance is associated with a **higher Z-score**.

The effect survives controls for major macroprudential tools.

Cost efficiency itself also enters **positively and significantly** throughout.

Dynamic Evidence: the average effect is horizon-dependent



Key message

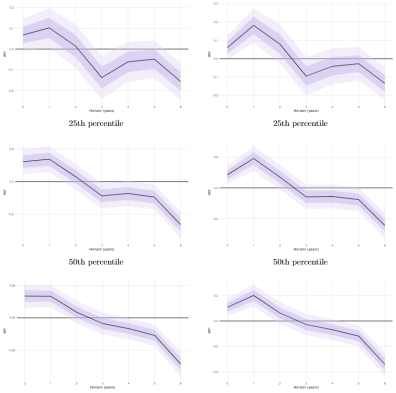
A tightening improves bank stability **on impact**, but the Z-score response **reverses at medium horizons**.

P75 banks show the smoothest path.

P25 banks revert earlier and more sharply.

Efficiency therefore shapes not just the *level* of resilience, but the **timing** of the response.

The dynamic pattern survives macroprudential controls



Once liquidity tools and LFX/LTV controls are added, the shape of the response remains: **initial stabilisation, later erosion**.

This supports the interpretation that the monetary policy effect is not merely proxying for macroprudential action.

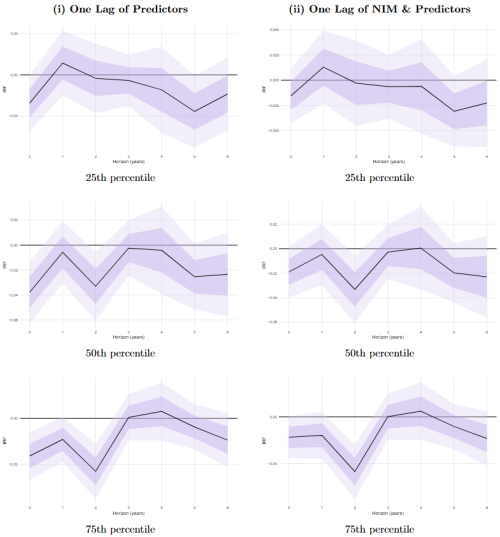
Channel evidence I: tightening compresses intermediation margins

	Hybrid policy		Official policy	
Specification	Contemp.	Lagged ctrls	Contemp.	Lagged ctrls
Baseline NIM regression	**-0.0368*** **	**-0.0344*** **	**-0.0369*** **	**-0.0364*** **
+ Liquidity & LFX	-0.0442***	-0.0460***	-0.0443***	-0.0473***
+ Liquidity & LTV	-0.0445***	-0.0453***	-0.0447***	-0.0468***

Monetary tightening **compresses NIMs** consistently across all specifications.

This margin compression is one reason the initial stability gain can fade over time.

Channel evidence II: NIM compression is most persistent for efficient banks



Interpretation

Efficient banks manage risk better, but they also experience more persistent margin compression after tightening.

The paper interprets this as a repricing / balance-sheet adjustment cost.

Better-managed banks smooth stability *despite* weaker NIM dynamics.

Putting the empirical story together

Three empirical facts

1. Tightening **raises Z-scores on impact**.
2. The gain **erodes at medium horizons** as borrower distress builds.
3. The deterioration is **smoother for high-efficiency banks**, even though their NIM compression can be more persistent.

Economic reading

Cost efficiency appears to proxy for the organisational ability to screen, monitor, reprice, provision, and absorb loan losses. That operational capability buffers the medium-run stability reversal.

Why move to a DSGE model?

Reduced-form evidence shows **sign reversal** and **heterogeneous smoothing**.

The model is built to explain both with two forces:

- a **fast risk-management channel**
- a **slow borrower-distress channel**

Bank types differ by cost efficiency, which shifts monitoring effectiveness and operational response.

Model intuition

Impact: tighter policy induces stronger monitoring and better screening, improving stability.

Medium run: debt-service stress and rising defaults dominate, pulling stability down.

Calibration and fit of the structural model

Parameter	Description	Value	Source
A. New Keynesian core			
β, σ, κ	discount, IES^{-1}, NKPC slope	0.99, 1.5, 0.10	Standard
ρ_R, ϕ_π, ϕ_y	Taylor rule	0.80, 1.50, 0.10	Standard
ρ_c	shock persistence	0.50	Calibrated
$\sigma_{\mathcal{U}}$	shock size	0.15 $\to$ 0.05	MD est.
B. Borrower technology			
δ_b	amortisation rate	0.15	Calibrated
η_R, η_y	debt-Euler semi-elast.	1.00, 1.00	Normalised
ρ_d	distress smoothing	0.85 $\to$ 0.96	MD est.
$\phi_{\Delta d}$	credit-crunch amplifier	0.10 $\to$ 0.44	MD est.
C. CSV default			
π_d	CSV elasticity	0.50 $\to$ 0.99	MD est.
λ_{LGD}	loss given default	0.45	HGC
χ_h	monitoring-vol elast.	0.38, 0.40, 0.55	Calibrated; robust $\pm 20\%$
D. Risk-management technology			
ρ_m	adj-cost ratio	0.40	Calibrated; robust $\pm 20\%$
ν_k	monitoring loading	0.35, 0.55, 0.80 $\to$ 1.32, 3.83, 3.32	MD est.
$c_{m,k}$	cost slope	0.05	Calibrated
E. Loan-pricing FOC			
ω_p	markup$\times$LGD	0.50	Implied
ω_m	monitoring pass-through	0.05	Calibrated
ω_c	capital-constraint feedback	0.25 $\to$ 0.21	MD est.
$\rho_{L,k}$	loan-rate stickiness	0.30, 0.35, 0.40	Panel NIM persistence
F. Deposit pricing (Drechsler et al. 2017)			
$\beta^D_{D,k}$	deposit markdown	0.92, 0.94, 0.95	DSS; panel confirms ordering
G. Equity accumulation			
ρ_e	equity persistence	0.90	Standard
ϕ_e	flow loading	0.15	$(1-\delta_{\mathrm{div}})\pi^H/\bar{\kappa}$
$\bar{\kappa}$	SS capital ratio	0.10	Basel III
H. Lending FOC			
ρ_l	portfolio stickiness	0.60	Calibrated
a_e, a_y, a_R	equity, output, policy elast.	0.50, 0.05, 0.00	Calibrated
$a_{r,k}$	own-rate semi-elast.	0.15, 0.25, 0.35	Calibrated
$a_{m,k}$	monitoring $\to$tightening	0.05, 0.10, 0.15 $\to$ 0.00, 0.00, 0.10	MD est.
I. Z-score decomposition			
$w_{\mathrm{NIM}}, w_{\mathrm{loss}}, w_{\mathrm{cost}}, w_{\mathrm{CAR}}$	ROA weights	0.15, 1.80, 0.00, 0.10	ΔZ decomposition
$w_{\mathrm{vol},k}$	vol channel	0.50, 0.60, 0.70 $\to$ 0.10, 0.19, 0.42	MD est.
J. NPL hazards			
$\rho_{npl,k}$	NPL persistence	0.80, 0.70, 0.65	Panel AR(1) range
$\iota_d, \iota_{m,k}$	inflow loadings	0.10, (0.22, 0.15, 0.12)	Panel; calibrated

The model is disciplined by matching local-projection moments.

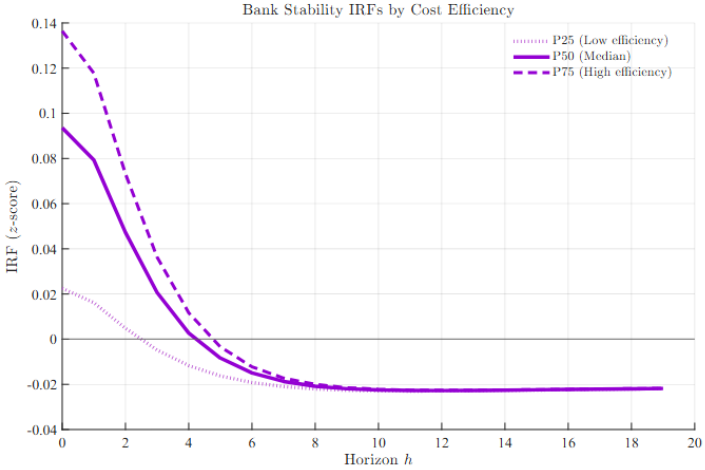
Weighted sum of squared errors falls by **45.8%** relative to the initial calibration.

The fitted model reproduces:

- the impact rise in Z-score,
- the medium-run reversal,
- NIM compression,
- heterogeneous credit responses.

Reported sign reversals occur around $h = 3$ **for P25** and $h = 5$ **for P50/P75**.

DSGE result 1: bank stability IRFs

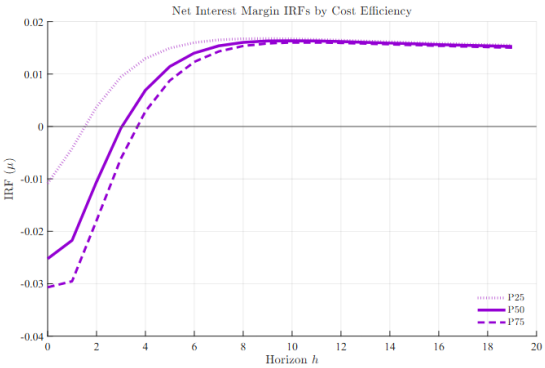


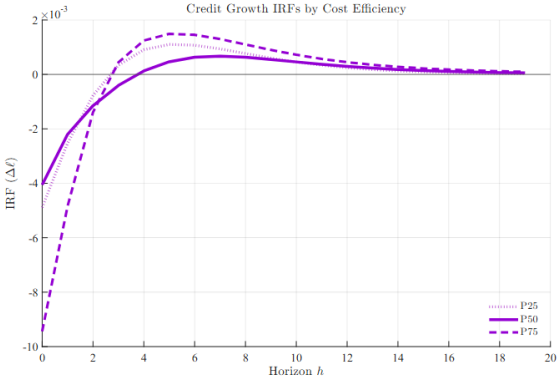
The model mirrors the empirical ranking: **P75 improves most on impact**.

All types later experience deterioration.

The least efficient banks reverse earliest.

DSGE result 2: NIM and credit growth

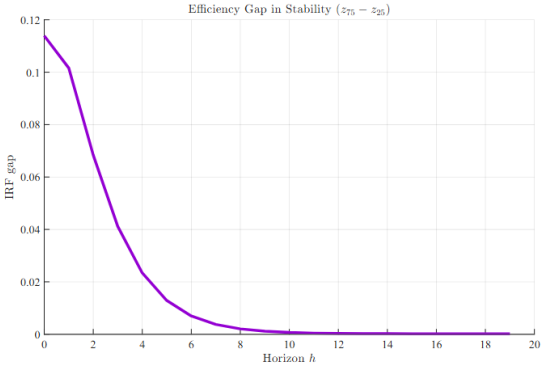


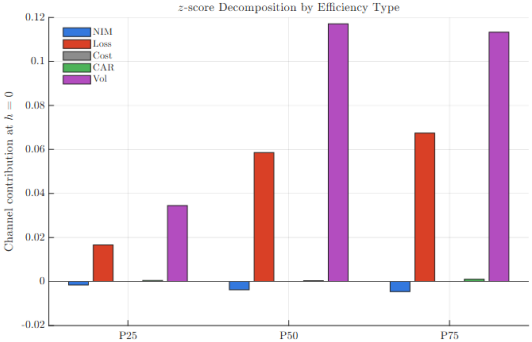


The model reproduces **NIM compression on impact**.

Credit growth responses are **heterogeneous**: the lending contraction is strongest for the most efficient banks in the fitted model.

DSGE result 3: efficiency gap and structural decomposition





The efficiency gap widens on impact and then closes over time.

The main stabilising forces are the **credit-loss reduction** and **volatility-reduction** channels.

The flow-margin channel is negative, but too small to overturn the impact stabilisation.

Counterfactual Experiments

Experiment	$z_{25}(0)$	$z_{50}(0)$	$z_{75}(0)$	$z_{75} - z_{25}$
(1) Baseline	$+0.0106$	$+0.0432$	$+0.1026$	$+0.0920$
(2) No deposit market power $\beta_{D,k} = 1$	$+0.0095$	$+0.0424$	$+0.1019$	$+0.0924$
(3) Homogeneous efficiency (avg parameters)	$+0.0483$	$+0.0483$	$+0.0483$	**0.0000**
(4) No risk-management channel $\nu_k = 0$	-0.0317	-0.0326	-0.0334	-0.0017
(5) Higher capital requirement $\bar{\kappa} = 0.15$	$+0.0106$	$+0.0432$	$+0.1027$	$+0.0920$

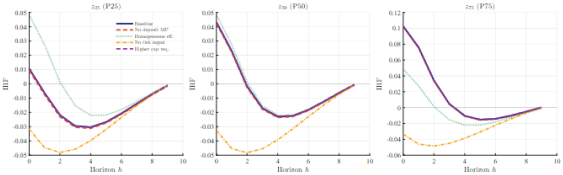
Homogeneous efficiency collapses the cross-bank stability gap to zero.

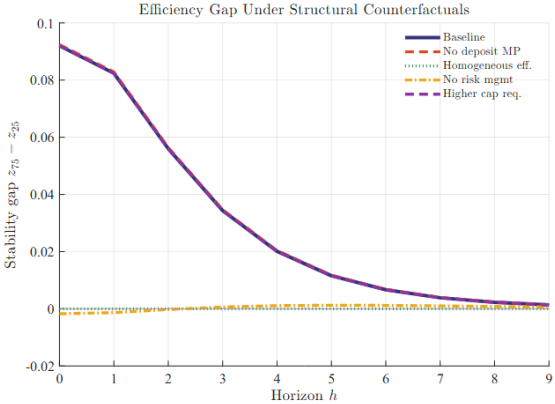
No risk-management channel turns the impact response negative for all bank types.

Removing deposit market power deepens NIM compression but changes stability only modestly.

Higher capital requirements have limited near-term effects in the linearised setup.

Counterfactual IRFs





The full paths confirm that **efficiency heterogeneity is structural**, not cosmetic.

Once heterogeneity is removed, the stability wedge disappears.

What should central banks take from this?

Policy implication 1

Tightening is neither universally stabilising nor universally destabilising. The banking effect is **state-dependent and horizon-dependent**.

Policy implication 2

Supervision should treat **cost efficiency as information about risk-control capacity**, not merely as an operating-performance score.

Policy implication 3

Macroprudential design should account for **efficiency heterogeneity**: banks with similar capital and liquidity positions may still react very differently to the same policy tightening.

Relevance for a Bank of Ghana audience

When inflation control requires tighter policy, the short-run banking response may look reassuring — but the paper warns against assuming the gains are permanent.

Monitoring should focus especially on **low-efficiency banks**, where the medium-run reversal is sharper.

A monetary tightening episode is more safely absorbed when supervisory policy strengthens:

- governance and internal controls,
- monitoring and provisioning technology,
- timely repricing and risk-recognition capacity.

The broad lesson is that **operational resilience and monetary transmission are linked**.

Conclusion

Bottom line

Monetary tightening tends to improve bank stability **on impact**, but that effect is **not monotone over time**. The medium-run reversal is materially shaped by **bank cost efficiency**.

Empirical evidence: FE, IV, and LP all point in the same broad direction.

Structural evidence: a DSGE model with heterogeneous banks reproduces the sign reversal and the smoothing effect.

Policy message: strengthening the **operational foundations of intermediation** can make monetary tightening safer from a financial-stability perspective.

Thank you.